

## Exercises for session 1: Static optimization

### Constrained and unconstrained optimization in the plane

Consider the following problem:

$$\min_{x,y} h(x,y) \text{ where } h(x,y) = (x-1)^2 + b(y-e^x)^2.$$

Here,  $b = 10$  is a parameter.

**Question 1:** Identify analytically the minimum point. Plot a contour plot of the function and mark the minimum point.

**Question 2:** Solve the optimization problem numerically using your favorite optimizer.

**Question 3:** Solve the optimization problem numerically, subject to an equality constraint. That is, define for  $s > 0$

$$J(s) = \min\{h(x,y) | x^2 + y^2 = s\}.$$

Determine  $J(1)$  numerically. Identify the associated Lagrange multiplier  $\mu$ . Verify numerically that

$$J(1 + \epsilon) \approx J(1) - \mu\epsilon,$$

taking, e.g.,  $\epsilon = 10^{-2}$ .

### A parametric program

We consider an optimization problem over  $u \in [0,1]$  in which a parameter  $x$  enters. Specifically, consider the function  $J : \mathbf{R} \mapsto \mathbf{R}$  given by

$$J(x) = \min_{u \in [0,1]} f(x,u) \text{ where } f(x,u) = (x-u)^2.$$

Also, define the optimizing  $u$ :

$$\mu(x) = \text{Arg} \min_{u \in [0,1]} f(x,u).$$

**Question 4:** Compute  $J(x)$  and  $\mu(x)$  and sketch or plot them. Verify that

$$J'(x) = \frac{\partial f}{\partial x}(x, \mu(x)).$$

*Optional:* Show that this last result holds not just for this particular parametric program, but in generality under weak regularity conditions.

### Completion of the square

Consider the quadratic functional in  $x, y$

$$f(x,y) = \frac{1}{2} \begin{pmatrix} x^\top & y^\top \end{pmatrix} \begin{bmatrix} A & B \\ B^\top & C \end{bmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

where  $x \in \mathbf{R}^n$ ,  $y \in \mathbf{R}^m$ , and the matrices have compatible dimensions. We take  $A = A^\top$  and  $C = C^\top > 0$ .

**Question 5:** Determine analytically

$$g(x) = \min_y f(x, y)$$

as well as the minimizing argument  $y$  as a function of  $x$ . Finally, determine  $f(x, y) - g(x)$ .

**Question 6:** Consider  $A = -1$ ,  $B = 1$ ,  $C = 1$ . Plot, as a function of  $x$ ,  $f(x, y)$  for 20 suitably chosen values of  $y$ . Include  $g(x)$  in the plot.

Plot also a contour plot of  $f(x, y)$  and indicate, for each  $x$ , the optimal  $y$ .

**Question 7:** Generate random matrices  $A = A^\top$ ,  $B$  and  $C > 0$  for, say,  $n = 20$  and  $m = 20$ ; use a probability distribution of your own choice. Solve the optimization problem  $\min_y f(x, y)$  numerically and verify that the solution agrees with analytical result. Changing the dimension or the distribution, do some quick experimentation to gain some experience with when the numerical solution agrees.

## Pareto optimality

The following is a simple example where two conflicting objectives must be traded off: An investor can buy several assets (e.g., stocks) and is interested in a large return. However, returns are random, and assets with high expected return also have high variance. The investor wants to maximize the expected return, but also to minimize the variance of the return.

Let  $X_i$  be independent random variables, each distributed as  $X_i \sim N(i, i)$ , for  $i = 1, 2, 3$ . We form a convex combination of these variables:

$$Y(w) = \sum_{i=1}^3 w_i X_i$$

where  $w \in \mathbf{R}^3$  are weights such that  $w_i \geq 0$ ,  $\sum_i w_i = 1$ .

Now, we have the two objectives, which ideally should both be small: The expected loss

$$J_1(w) = -\mathbf{E}Y(w),$$

and the variance on the loss

$$J_2(w) = \mathbf{V}Y(w).$$

**Question 8:** Find the  $w$  that minimizes  $J_1(w)$ . Next, find the  $w$  that minimizes  $J_2(w)$ .

**Question 9:** Determine the Pareto front. Specifically, for various values of  $\theta \in [0, 1]$ , solve the problem

$$\min_w [\theta J_1(w) + (1 - \theta) J_2(w)]$$

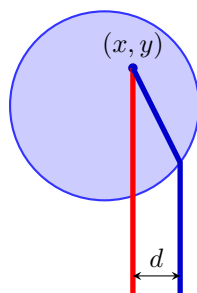
numerically, subject to the constraints given. Plot the front in the  $(J_1, J_2)$ -plane. Plot also the optimal weights  $(w_1, w_2, w_3)$  as functions of  $\theta$ . Can you explain qualitatively the shape of the curves between the end points?

## Maximizing the effect of refraction

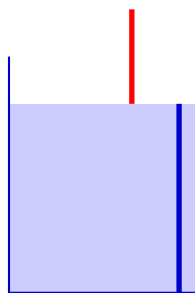
**Question 10:** A thin straw is placed vertically in a circular cylindrical glass with radius 1, which is half full of water. When viewed from the side, it appears that the upper and lower parts of the straw are a distance  $d$  apart; see the figure. Choose the position  $(x, y)$  of the straw which maximizes the distance  $d$ . Use a refractive index of 1.33 for water.

*Note:* This exercise can be approached in many different ways and it is worthwhile to at least consider the options.

Top view



Side view



## Convexity

**Question 11:** Which of the optimization problems considered in the previous are convex? Does it matter?